

Multiply generated hypercyclic algebras: the question in arXiv:1710.10413 was answered by arXiv:1706.08651

Literature-status packet for run `fa_banach_001`

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Status: literature already answered affirmatively. The source says that the existence of multiply generated algebras of hypercyclic entire functions was unknown. Bès–Papathanasiou give a countably generated *free dense* hypercyclic algebra for the differentiation operator, a much stronger existential answer. This is a literature identification, not a new construction from the present run.

Original question

Luis Bernal-González, J. Alberto Conejero, George Costakis, and Juan Seoane-Sepúlveda, *Multiplicative structures of hypercyclic functions for convolution operators*, arXiv:1710.10413v1 [1], state on source-PDF page 3:

So far, the existence of multiply generated algebras of hypercyclic entire functions is unknown.

Here a hypercyclic algebra for an operator T on the Fréchet algebra $\mathcal{H}(\mathbb{C})$ is a subalgebra A such that $A \setminus \{0\} \subseteq \text{HC}(T)$. The source then obtains an infinitely generated *multiplicative group* G for subexponential convolution operators $\Phi(D)$, all of whose elements except 1 are hypercyclic. It notes that the algebra generated by G is infinitely generated, but does not claim every nonzero sum in that algebra is hypercyclic. Thus the group theorem does not itself answer the algebra question.

Independent decisive answer

Juan Bès and Dimitris Papathanasiou, *Algebrable sets of hypercyclic vectors for convolution operators*, arXiv:1706.08651 [2], formulate on source-PDF page 2 the precise question:

Is it possible to find a hypercyclic algebra for D that is not finitely generated?

They immediately state that they answer it affirmatively. Their Theorem 1.1 proves, under a geometric level-set condition on a convolution symbol Φ , that a generic sequence in $\mathcal{H}(\mathbb{C})^{\mathbb{N}}$ freely generates a dense hypercyclic algebra for $\Phi(D)$. Corollary 1.3 specializes this to the MacLane differentiation operator D :

$\text{HC}(D) \cup \{0\}$ contains a dense free algebra on countably many generators.

This is strictly stronger than merely finding a multiply generated hypercyclic algebra: the generators are algebraically independent, there are countably many of them, and the resulting algebra is dense in $\mathcal{H}(\mathbb{C})$.

Identification, chronology, and scope

The match is exact at the existential level. The source’s unqualified question asks whether any multiply generated hypercyclic algebra of entire functions exists; the independent paper constructs one for D , itself a convolution operator, and explicitly says it answers Aron’s equivalent not-finitely-generated formulation. The supporting arXiv record was first submitted in June 2017, whereas arXiv:1710.10413 was submitted in October 2017; its proof-bearing version was subsequently revised in March 2019. The chronology explains why this is best viewed as an independent literature answer rather than a new result of this run.

The answer does not say that *every* nonconstant subexponential symbol Φ supports such an algebra. That stronger universal question is not what the quoted sentence asks, and the source itself only promises a multiplicative group for arbitrary such Φ . Bounded searches of the exact quoted phrase, “not finitely generated hypercyclic algebra,” “free hypercyclic algebra,” the source title, and both arXiv identifiers found the Bès–Papathanasiou paper as the decisive direct match. The packet stores both PDFs locally.

References

- [1] L. Bernal-González, J. A. Conejero, G. Costakis, and J. Seoane-Sepúlveda, *Multiplicative structures of hypercyclic functions for convolution operators*, J. Operator Theory 80 (2018), 213–228; [arXiv:1710.10413](#).
- [2] J. Bès and D. Papathanasiou, *Algebrable sets of hypercyclic vectors for convolution operators*, [arXiv:1706.08651](#), Theorem 1.1 and Corollary 1.3.